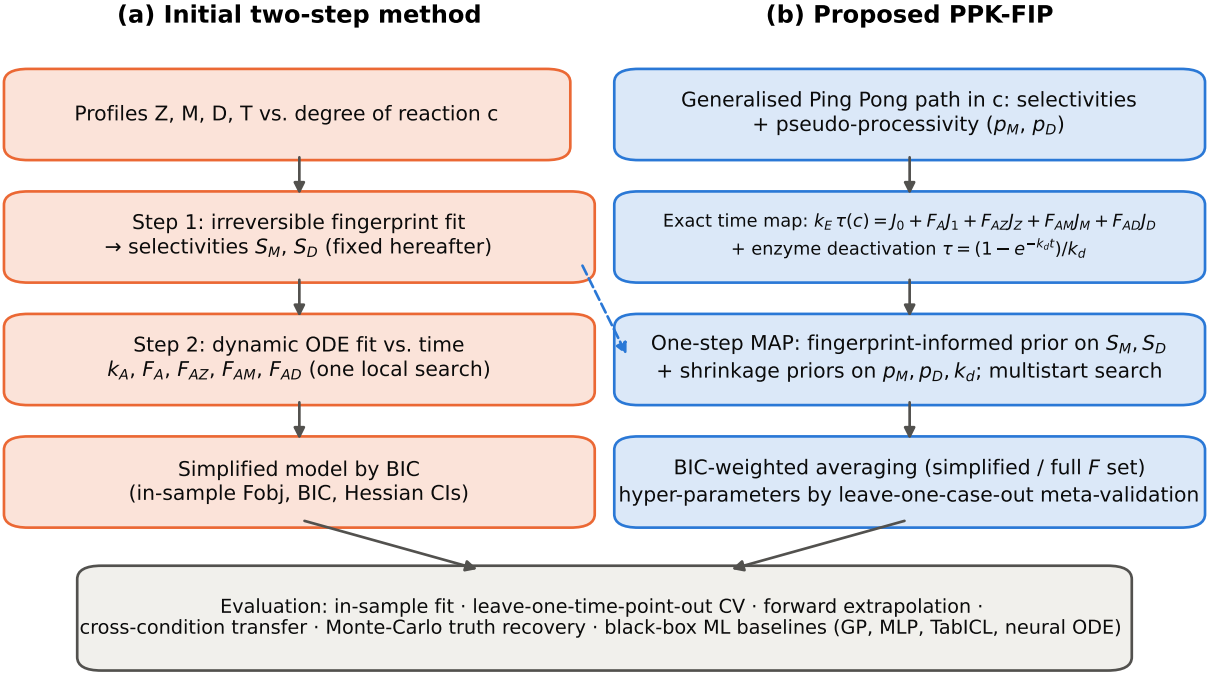
(a) Initial two-step method
(b) Proposed PPK-FIP
Profiles Z, M, D, T vs. degree of reaction c
Step 1: irreversible fingerprint fit
→ selectivities S_M, S_D (fixed hereafter)
Step 2: dynamic ODE fit vs. time
k_A, F_A, F_{AZ}, F_{AM}, F_{AD} (one local search)
Simplified model by BIC
(in-sample Fobj, BIC, Hessian CIs)
Generalised Ping Pong path in c: selectivities
+ pseudo-processivity (p_M, p_D)
Exact time map: $k_E\,\tau(c) = J_0 + F_A J_1 + F_{AZ} J_Z + F_{AM} J_M + F_{AD} J_D$
+ enzyme deactivation $\tau = (1 - e^{-k_d t})/k_d$
One-step MAP: fingerprint-informed prior on S_M, S_D
+ shrinkage priors on p_M, p_D, k_d; multistart search
BIC-weighted averaging (simplified / full F set)
hyper-parameters by leave-one-case-out meta-validation
Evaluation: in-sample fit · leave-one-time-point-out CV · forward extrapolation ·
cross-condition transfer · Monte-Carlo truth recovery · black-box ML baselines (GP, MLP, TabICL, neural ODE)